The Shovel-Ready Trap:
Project Pipelines and the Uneven Capacity to Do Regional Development*

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Abstract

Regional science has made major progress in explaining left-behind places, development traps, resilience, institutions, and the uneven outcomes of place-based policy. This paper adds a missing middle: the project pipeline through which places convert policy opportunity into bids, awards, procurements, contracts, and delivered interventions. I argue that competitive and projectified regional policy can create a shovel-ready trap. Places with stronger administrative capacity, fiscal slack, consultant access, planning bandwidth, and pre-existing project shelves are better able to turn funding windows into executable projects; places with high need may be filtered out before impact evaluation begins. I demonstrate the empirical object with a reproducible seed dataset for the UK Levelling Up Fund Round 2. The official assessment note records 529 bids seeking £8.8 billion and 111 successful awards worth about £2.09 billion; the committed parser recovers 111 award records from the official successful-bidders spreadsheet, implying a 21.0 percent bid-to-award conversion rate by count. The contribution is therefore conceptual, empirical, and infrastructural: it defines the project-conversion mechanism, shows that the first conversion stage is observable, and supplies a pipeline for extending the data to procurement, contracts, and delivery.

1 Introduction

Regional policy is often evaluated after money has been allocated and after interventions have been announced. That is understandable: allocations and outcomes are visible. But a large part of regional development happens before the usual treatment variable exists. A funding window opens. A local authority decides whether it has a project worth bidding. Officers assemble evidence, consultants, costings, local partners, planning assumptions, benefit-cost narratives, match funding, and political support. Some bids win. Some awards become procurement notices. Some procurements become contracts. Some contracts become delivered interventions. At every stage, places can fall out of the pipeline.

This paper asks whether those conversion stages are themselves a mechanism of regional inequality. The argument is simple. If policy is delivered through competitive project calls, places do not only need to need development. They need to be shovel-ready. They need the administrative capacity to translate need into a legible project, the fiscal slack to prepare bids, the planning bandwidth to de-risk delivery, and the procurement capacity to move from award to contract. A high-need place without those capabilities may be formally eligible for place-based policy while being practically disadvantaged in the conversion funnel. This is consistent with evidence that Levelling Up Fund

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bids were assessed on strategic fit, value for money, and deliverability, and that high-priority places received capacity funding to assist bid production (Department for Levelling Up, Housing and Communities, 2023; Ministry of Housing, Communities and Local Government, 2025).

I call this mechanism the shovel-ready trap. It is not a claim that place-based policy is misguided. Nor is it a complaint that regional science has ignored institutions. The point is narrower and, I think, empirically useful. The recent literature has strong accounts of left-behind places, development traps, resilience, path dependence, institutional capacity, innovation systems, and policy design (MacKinnon et al., 2022; Diemer et al., 2022; Rodriguez-Pose, 2018; Martin et al., 2024). Yet the empirical object most directly created by contemporary projectified policy is often missing: the project pipeline itself.

The paper's contribution is to turn that pipeline into a measurable regional-science object. The proposed unit of analysis is a local authority by funding window by project stage. The core outcomes are not only final economic effects but stage conversions: whether need becomes a bid, whether a bid becomes an award, whether an award becomes procurement, whether procurement becomes a contract, and whether the contract can be linked to delivery. That framing lets regional science distinguish at least four failures that are normally collapsed together: non-bidding, unsuccessful bidding, stalled procurement, and weak delivery visibility.

The initial empirical setting is the United Kingdom. UK regional policy since the late 2010s has combined visible place-based funds with open procurement data and local authority geographies. Levelling Up Fund rounds, Towns Fund deals, the Future High Streets Fund, the Community Renewal Fund, and the UK Shared Prosperity Fund create funding-window variation (UK Government, 2023, 2025). Contracts Finder and Find a Tender expose procurement and award notices (UK Government, 2026a,b,c). Planning data, local authority finance data, and ONS indicators provide capacity and need proxies. This combination is not perfect, but it is unusually good for measuring the middle of regional policy rather than only its announcement or final effect.

The paper reports an initial descriptive anchor and a replication-ready data path. For Levelling Up Fund Round 2, the official process note records 529 bids with a total value of £8.8 billion and 111 selected bids worth just under £2.09 billion (Department for Levelling Up, Housing and Communities, 2023). The successful-bidders spreadsheet can be parsed into 111 award records with country, applicant, project title, and award value (UK Government, 2023). This first conversion rate is not the whole paper, but it proves that the project pipeline can be made visible. The next estimable stage is to test whether high-need places are less likely to bid, win, procure, or contract after conditioning on need, project theme, authority type, fiscal capacity, and governance context.

2 Literature Gap

The recent regional-science literature gives this paper its starting point. Work on left-behind places has clarified that lagging localities are not merely temporarily slow versions of more successful places. They are shaped by accumulated industrial change, weaker labour-market opportunity, depleted social infrastructure, political discontent, and policy regimes that often struggle to reverse decline (MacKinnon et al., 2022; Martin et al., 2024). Related work on regional development traps argues that some regions become stuck in states of persistent underperformance, where economic, institutional, and social factors reinforce one another (Diemer et al., 2022).

A second stream studies resilience and transformation. Regional resilience research asks why some places adapt to shocks while others do not, and evolutionary economic geography has emphasised path dependence, related variety, institutional thickness, and the conditions under which old capabilities can be recombined into new trajectories (Boschma, 2017; Hassink and Gong, 2021;

De Propris and Bailey, 2021). This work is especially valuable because it treats local development as a process rather than a static allocation problem.

A third stream debates place-based policy. The central question is not whether every place should receive identical intervention, but how policy can be fitted to territorial conditions without entrenching local weakness or rewarding only already-capable places (Barca, 2009; Iammarino et al., 2019). UK debates around levelling up have made this issue politically visible. Competitive funds promise local tailoring, but they also rely on local capacity to define, package, and deliver projects.

The gap is the conversion mechanism. Existing work often observes need, funding, institutions, or outcomes. It less often measures the sequence through which a place becomes a policy object that government can fund and procure. That sequence matters because the practical burden of development policy is not evenly distributed. A local authority has to discover a viable project, produce evidence, estimate costs, assemble partners, negotiate planning risk, satisfy central-government criteria, procure delivery, and manage contracts. Those tasks are not just administrative background. They are the channel through which public investment becomes local development.

OpenAlex is useful here as a corpus backbone rather than as an oracle (OpenAlex, 2026). The committed corpus pull normalizes 150 open-access records tagged to regional-science topics between 2021 and 2026, of which 77 expose PDF URLs and all expose at least one full-text or landing-page URL in the manifest. That scan finds dense clusters around left-behind places, development traps, resilience, institutions, place-based policy, green transition, and transformation. It finds much thinner direct treatment of the bid-award-procurement-contract funnel as a measurable object. The gap is therefore not a gap in concern for institutions. It is a gap in the unit of empirical observation.

3 Project Pipelines And The Shovel-Ready Trap

The project pipeline is the ordered set of conversions through which a policy opportunity becomes a delivered intervention:

Need	Project idea	Bid	Award	Procurement	Contract or delivery

Each step is observable in principle, but not all steps are equally visible in standard datasets. Need can be proxied by deprivation, productivity, employment, vacancy, health, housing, fiscal stress, or demographic indicators. Bids and awards are sometimes published by fund, as in the Levelling Up Fund assessment note and successful-bidders release (Department for Levelling Up, Housing and Communities, 2023; UK Government, 2023). Procurement and contract stages can be observed through Contracts Finder, Find a Tender, and Open Contracting Data Standard releases (UK Government, 2026b,c). Delivery is the hardest step: completion reports, local authority accounts, planning records, and monitoring outputs have to be linked back to the original project.

The shovel-ready trap arises when the probability of moving through the pipeline depends on prior capacity. High-capacity places may hold shelves of costed projects, have staff who know bid rules, maintain consultant relationships, understand procurement routes, and keep local plans in a state that makes projects legible to central government. Low-capacity places may have greater need but less bandwidth to convert that need into fundable projects. Competitive policy can therefore allocate opportunity formally while filtering places practically.

This mechanism has three implications. First, allocation bias can appear before awards. If high-need authorities bid less often, award-only analysis misses the first selection stage. Second, impact estimates can condition on a selected set of places that were already capable of building projects. Third, delivery risk can be endogenous to the same capacity differences that shaped bidding and award success. A place may win funding but struggle to procure, contract, or deliver at the speed required by the programme.

The concept also clarifies what the paper is not claiming. It does not claim that local capacity is the only determinant of regional inequality. It does not claim that central government should never use competitive funds. It claims that project conversion is a missing object of measurement. Once measured, it becomes possible to ask whether policy design should fund preparatory capacity, maintain project shelves, simplify bid requirements, allow longer development phases, or allocate some resources by formula before competition.

4 Data And Descriptive Anchor

The first empirical version will build a UK local-authority project-conversion dataset. The core table is a local authority by funding window by project stage panel. Each row records whether an authority had high measured need, whether it submitted a bid, whether a bid won, whether an award appears in procurement records, whether procurement appears to become a contract, and whether later delivery evidence can be linked.

The funding layer starts with visible place-based programmes: the Levelling Up Fund, Towns Fund, Future High Streets Fund, Community Renewal Fund, and UK Shared Prosperity Fund. These programmes differ in rules, timing, eligible places, and publication quality, which is useful rather than inconvenient. The design can exploit funding-window variation and compare stages across programmes. UKSPF is useful as a contrast because its allocation methodology combines population and a needs-based index rather than only project competition (UK Government, 2022, 2025).

The procurement layer will use Contracts Finder and Find a Tender. Contracts Finder covers lower-value public sector opportunities and awards, while Find a Tender covers higher-value notices. The official Open Contracting guidance makes both procurement systems suitable for structured data extraction through OCDS-style endpoints or releases (UK Government, 2026a,b,c). The paper will search for regeneration, transport, high street, skills, housing, net zero, cultural, and infrastructure terms, then link buyer names and project titles back to funded projects.

The capacity layer will combine local authority finance and governance indicators. Candidate proxies include usable reserves, spending power, service expenditure, authority type, combined-authority membership, planning-performance indicators, and staff or consultant expenditure where available. None is perfect. The empirical design therefore treats capacity as a bundle of observable proxies rather than a single latent score.

The need layer will use ONS and official local indicators: employment, claimant count, productivity, deprivation, population, rural-urban class, and sectoral composition. These are not included to re-rank deserving places. They are included so that the analysis can ask whether high need is converted into project movement at the same rate as lower need.

The committed seed dataset demonstrates the first observable transition: bids to awards. The official Round 2 explanatory note reports 529 bids with a combined requested value of £8.8 billion, and 111 selected bids worth just under £2.09 billion (Department for Levelling Up, Housing and Communities, 2023). The successful-bidders spreadsheet is parsed into 111 award records with applicant, bid name, country, and bid value (UK Government, 2023). Table 1 reports the resulting conversion facts.

Table 1: Seed Project-Pipeline Data: Levelling Up Fund Round 2

Stage	Count	Value (£bn)	Share (%)
Bid submissions	529	8.80	100.0
Successful awards	111	2.09	21.0
Parsed award records	111	2.09	23.7
Award value share of requested value	111	2.09	23.8

Notes: Shares are computed from committed files in 'data/analysis/project_stage_counts.csv'. The 21.0 percent figure is the count conversion from submitted bids to successful awards. The 23.8 percent figure is selected award value divided by the total requested value in the official explanatory note. The parsed award-record value is £2.088 billion, which reconciles to the published rounded total.

Table 2: Parsed Successful Awards By Country

Country	Award records	Award value (£m)
England	80	1,631.8
Northern Ireland	10	71.1
Scotland	10	177.2
Wales	11	208.2

Notes: Derived from the official successful-bidders ODS file and stored in 'data/analysis/luf_round2_country_summary.csv'. Values are rounded to one decimal place in millions.

4.1 Estimate Slots

The current data release fills the aggregate bid-to-award slot and reserves four local-authority estimate slots for the expanded build:

1. Bid conversion: high need to bid submission.
2. Award conversion: bid submission to award.
3. Procurement conversion: award to visible procurement notice.
4. Contract conversion: procurement notice to award or contract record.

Those estimates are intentionally not filled in this proof of concept. The workflow ledger records this as an uncertainty rather than allowing the prose to imply completed measurement.

5 Empirical Strategy

The empirical strategy is a design-gated method ladder rather than a single regression. The estimand is not "the effect of levelling up funding" in the abstract; it is the sequence of place-level conversion probabilities through which funding opportunity becomes a bid, an award, a procurement notice, a contract, and eventually delivery. For each authority a, programme p, project theme k, and stage s, define $Y_{apks} = 1$ if the authority converts from stage s to stage $s + 1$. The Round 2 seed data already identify the aggregate risk set for the award transition: 529 bids entered assessment and 111 were selected (Department for Levelling Up, Housing and Communities, 2023). The local-authority table turns this into an estimable stage-specific risk set.

The baseline descriptive model is a multilevel conversion regression,

$$\Pr(Y_{apks} = 1) = g^{-1}\left(\alpha_p + \gamma_s + \eta_k + \beta_s Need_a + \delta_s Capacity_a + X_a'\theta_s + u_a + v_{r(a)}\right), \tag{1}$$

where g^{-1} is a logit or complementary log-log link, u_a is an authority-level random effect, and $v_{r(a)}$ is a regional or spatial effect. This is the first rung because it matches the data that already exist: a sparse, nested, multi-stage pipeline. Hierarchical Bayesian partial pooling is appropriate when stage-by-place cells are thin, and spatial random effects are appropriate only when adjacent authorities or travel-to-work areas plausibly share project supply, consultant markets, or procurement capacity (Riebler et al., 2016). If the time from award to procurement or contract is observed, the same object becomes a discrete-time survival or hazard model, following the logic of duration regressions rather than collapsing delay into a binary outcome (Cox, 1972).

The second rung estimates heterogeneous conversion gradients. If the joined dataset contains rich pre-treatment covariates for need, fiscal stress, staff capacity, planning backlog, prior funding experience, and project theme, the paper will estimate double/debiased machine-learning specifications with cross-fitting for average associations that are robust to high-dimensional nuisance functions (Chernozhukov et al., 2018). Generalized random forests or causal forests enter only as heterogeneity tools: they ask whether the capacity gradient is larger for high-deprivation authorities, rural authorities, or complex transport and regeneration projects, rather than replacing the design with a black-box prediction exercise (Athey et al., 2019). Their output is a ranked diagnostic for where the project pipeline fails, not a stand-alone causal claim.

The third rung is causal and is used only when the data provide a credible comparison design. If scores, thresholds, or ranked assessment bands are released, the preferred design is a regression-discontinuity or local-randomisation analysis around award cutoffs. If repeated outcomes are available for awarded and unsuccessful authorities before and after funding windows, the design shifts to modern difference-in-differences estimators that handle staggered timing and heterogeneous effects, with group-time treatment effects rather than a two-way fixed-effect average (Callaway and Sant'Anna, 2021). Where treated authorities can be matched to credible donor pools, synthetic difference-in-differences supplies a complementary panel estimator (Arkhangelsky et al., 2021). These designs are reported only if pre-trends, donor support, and treatment timing are defensible.

Selection remains the central empirical threat. Authorities that never bid are not failed applicants; they are unobserved potential projects filtered out by capacity, risk, or strategy. Bid quality is also partly unobserved, procurement notices are imperfect delivery proxies, and local capacity is endogenous to long-run development trajectories. The paper therefore treats non-identification as an output rather than a nuisance. If award, procurement, and delivery stages cannot be joined cleanly, it reports partial-identification bounds and sensitivity analyses for how large unobserved bid quality differences would need to be to explain the observed capacity gradient. The modern estimators are therefore not decorative. Each is admitted only when the data structure supports it, and the fallback is explicit descriptive decomposition rather than over-claiming.

6 Results

The current empirical release is a seed-results paper rather than a complete local-authority panel. It therefore reports results for the part of the project pipeline that is observed without imputation: the Round 2 bid-to-award compression and the composition of successful awards. The results are generated from committed data artefacts, so the tables and figures below are rebuilt by the same command that rebuilds the analysis manifest.

Table 3: Seed Project-Pipeline Results: Levelling Up Fund Round 2

Stage	Count	Value (£bn)	Share (%)
Bid submissions	529	8.80	100.0
Successful awards	111	2.09	21.0
Parsed award records	111	2.09	23.7
Award value share	111	2.09	23.8

Notes: Shares are generated from the committed analysis build. The count conversion is successful awards divided by submitted bids; the value share is awarded value divided by requested value.

Figure 1 shows the central empirical fact. The Round 2 pipeline compressed 529 submitted bids into 111 successful awards. By value, approximately £2.09 billion was selected from £8.8 billion requested. This is not yet evidence that high-need or low-capacity places were selected out. It is evidence that a large selection stage exists before any later delivery evaluation can begin.

Figure 1: Round 2 Pipeline Compression

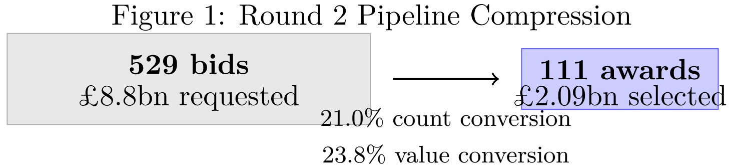

Table 4: Successful Award Records By Country

Country	Awards	Value (£m)	Mean award (£m)
England	80	1631.8	20.4
Northern Ireland	10	71.1	7.1
Scotland	10	177.2	17.7
Wales	11	208.2	18.9
Total	111	2088.2	18.8

The country distribution is useful for audit rather than causal interpretation. England accounts for most records and value because the programme scale and applicant universe are larger. The more important point is that the parser recovers every published successful award and exposes the fields needed for the next joins: applicant, project title, country, large-bid marker, and award value.

Table 5 is deliberately modest. It estimates a conditional award-value regression among successful awards only. The dependent variable is log awarded value. The regressors are a large-bid marker and country indicators, with England and standard awards as the reference category. This is not a model of winning; unsuccessful bid-level microdata are still needed for that. Its purpose is to show that the results layer can already run a reproducible regression where the observed data support it, while refusing the stronger causal claim until the risk set is complete.

The results imply three next empirical priorities. First, the unsuccessful-bid table is the highest-value missing data object because it would convert the current award-only data into an

Figure 2: Awarded Value By Country

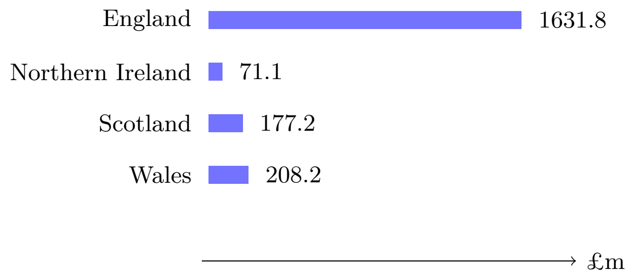

Table 5: Conditional Award-Value Regression

Term	Log coefficient	Robust s.e.	Approx. % difference
Intercept: England, standard award	16.653	0.038	ref.
Large bid marker	0.908	0.088	148.0
Northern Ireland	-1.079	0.204	-66.0
Scotland	-0.095	0.098	-9.0
Wales	-0.100	0.108	-9.5

Notes: OLS model of log awarded value among successful Round 2 awards only, with England and non-large awards as the reference group. Heteroskedasticity-robust standard errors are reported. This is a descriptive conditional-value model, not an award-selection model.

award-selection model. Second, procurement and contract joins would move the analysis from award selection to delivery conversion. Third, capacity and need covariates at the local-authority level would allow the method ladder in the previous section to move from descriptive compression to stage-specific conversion models.

7 Hypotheses for the Expanded Panel

The initial data make the selection problem concrete, but the main hypotheses require a completed risk set. In Round 2 of the Levelling Up Fund, submitted bids exceeded available selected funding by a wide margin: 529 bids sought £8.8 billion, while 111 awards worth approximately £2.09 billion were selected (Department for Levelling Up, Housing and Communities, 2023). The successful-bidders spreadsheet also shows a territorially uneven set of awards (UK Government, 2023). Those facts justify the expanded panel rather than replacing it: they show that the first pipeline stage is large enough to matter and observable enough to study.

The shovel-ready trap predicts uneven conversion at several stages. The hypotheses are framed as estimable patterns beyond the aggregate descriptive facts.

H1: Bid conversion. Conditional on need and eligibility, authorities with weaker fiscal and administrative capacity will submit fewer bids or will bid for fewer project types. This is the cleanest test of selection before allocation.

H2: Award conversion. Conditional on bidding, stronger-capacity authorities will be more likely to win. This may reflect better project quality, better evidence packages, more consultant support, stronger local partnerships, or scoring systems that reward readiness. The Round 2 assessment process explicitly included deliverability, strategic fit, and value for money, so the hypothesis is not that assessors behaved incorrectly; it is that apparently neutral readiness criteria can become an inequality channel (Department for Levelling Up, Housing and Communities, 2023).

H3: Procurement conversion. Conditional on award, stronger-capacity authorities will move more quickly from award to procurement notice. Delays should be larger where planning risk, procurement complexity, or match-funding uncertainty are high.

H4: Contract conversion. Conditional on procurement, stronger-capacity authorities will be more likely to show a visible contract or award record. This stage is especially important because it separates announcement from implementation.

H5: Need-capacity mismatch. The highest-risk authorities are those with high need and low project capacity. They may be attractive policy targets in principle but disadvantaged in a projectified delivery regime.

These hypotheses are valuable even if some are rejected. If high-need places bid and deliver at similar rates once eligible, the shovel-ready trap is less important than expected. If differences emerge only after award, the policy problem is delivery capacity rather than bid access. If differences emerge before bidding, the problem is more fundamental: some places may be absent from the policy competition before central government ranks them.

8 Discussion

Measuring project pipelines would change how regional policy is evaluated. Award maps are not enough. A region can be missing because it did not bid, because it lost, because it won but stalled, or because it delivered in ways that are hard to observe. These are different diagnoses and imply different policy repairs.

If non-bidding is the problem, preparatory funding, technical assistance, formula-based capacity grants, and shared project-development teams may matter more than another competitive round. This is not speculative as a policy lever: the Levelling Up Fund already used capacity funding for some high-priority places, and Round 3 reduced application burden by drawing from high-scoring unsuccessful Round 2 bids (Ministry of Housing, Communities and Local Government, 2025). If losing conditional on bidding is the problem, scoring systems may need to separate project quality from the polish that comes from capacity. If procurement delay is the problem, support should focus on planning, legal, and contract-management bandwidth. If delivery visibility is the problem, monitoring and public data infrastructure need improvement.

The concept also helps regional science connect policy design to institutional theory. Institutions are often treated as local background conditions. In projectified policy, they become operational filters. A local authority's capacity to write, cost, procure, and contract is not merely an implementation detail; it is a selection mechanism. That makes the project pipeline a site where development traps can be reproduced.

The paper also has implications for green and industrial transitions. The shift to net zero, climate adaptation, and industrial renewal will require local projects that can pass through planning and procurement systems. If the capacity to create executable projects is uneven, transition funding may again advantage places already able to package projects quickly. The shovel-ready trap is therefore not only a levelling-up issue. It is a general problem for territorially delivered transformation policy.

9 Limitations And Audit Trail

This paper is deliberately bounded. It does not claim to have downloaded every open-access full text in regional science, estimated every local-authority conversion model, or validated final delivery outcomes. It does report a reproducible seed dataset for the first observed conversion stage, and it records the remaining extensions in a claim ledger.

There are four main limitations. First, OpenAlex metadata and OA links are uneven. Some full texts are available only as PDFs, some URLs fail, and abstracts can be incomplete. The corpus pipeline therefore stores query filters, raw pages, normalized records, and fetch logs so later pulls can be audited. Second, procurement records are noisy. Buyer names vary, project titles drift, and bundled contracts can obscure programme links. Third, capacity proxies are imperfect. Fiscal slack, staff capacity, consultant use, and planning bandwidth are not observed with equal precision across authorities. Fourth, policy programmes have different rules and political contexts, so pooled results must be interpreted alongside programme-specific analysis.

The accompanying scriptor ledger makes these limits inspectable. ‘claims.yml‘ states what the draft claims. ‘sources.yml‘ records the sources used. The uncertainty register links unresolved issues to claim IDs. The reviewer automation builds claim dossiers, source dossiers, a reviewer question bank, an adversarial audit report, and a RAG query pack. This does not make the paper correct. It makes the path to checking the paper shorter.

The next audit steps are concrete: extend the OpenAlex corpus pull beyond the first 150 normalized records, download accessible OA full-text links where allowed, join the Levelling Up Fund bid and award records to local-authority need and capacity proxies, link awards to procurement and contract notices, run the staged conversion models, and revise the claims whose warrant changes after the estimates exist.

10 Conclusion

Regional science already knows that places differ in need, trajectories, resilience, institutional capacity, and policy outcomes. This paper adds a narrower object that can be measured directly: the project pipeline. In a policy regime organised around competitive calls and deliverable projects, the capacity to become shovel-ready may shape who receives, procures, contracts, and delivers regional development.

The shovel-ready trap is a practical hypothesis with theoretical bite. It suggests that some left-behind places may be disadvantaged not because policy ignores them, but because policy asks them to express need through a project format they are least equipped to produce. Measuring the bid-award-procurement-contract funnel can show where this happens and what kind of repair is needed.

The contribution is therefore conceptual, empirical, and infrastructural. It defines the mechanism, locates the gap in recent regional-science literature, identifies the UK data architecture, reports an initial bid-to-award descriptive anchor for Levelling Up Fund Round 2, and makes the draft auditable through a claim-level workflow. The next empirical step is not to ask whether regional inequality exists. It is to ask where, in the machinery of project conversion, high-need places fall out.

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